SARA IMRAN

LLM Systems Engineer

Karachi, Pakistan | sara.imran@example.com | Professional experience: 4 years

PROFESSIONAL PROFILE

LLM systems engineer with four years of professional experience designing search-grounded assistants, document-intelligence pipelines and measurable NLP services. Strong evidence of model integration, orchestration and evaluation, primarily with Haystack, DSPy and custom Python components rather than LangChain.

TECHNICAL SKILLS

- Python, FastAPI, PyTorch, scikit-learn, Hugging Face Transformers
- NLP, question answering, NER, document classification, dense retrieval and hybrid search
- LLM integrations using hosted model APIs; Haystack pipelines; DSPy prompt/program optimization
- Custom graph/state orchestration, manual approval gates, retries, validation, tools and routing
- Evaluation: recall@k, exact match, grounded-answer review, hallucination annotation, latency
- GCP Cloud Run, BigQuery, Docker, Postgres; time-series features and anomaly detection

PROFESSIONAL EXPERIENCE

Senior LLM Engineer | Atlas Searchworks (fictional) | **Oct 2024 - Present**

- Delivered a customer-facing knowledge assistant using Haystack document retrieval, hosted LLM chat APIs and a custom Python decision pipeline.
- Implemented tool routing, conditional branches, memory/session state, explicit escalation and source-linked answer generation without LangChain.
- Designed an independent 450-question grounded-answer audit: source coverage improved from 71% to 90% while answer abstention remained conservative.
- Built Python ingestion pipelines for PDFs and DOCX, including deduplication, metadata joins, relevance testing, and privacy-aware retention.
- Exposed inference endpoints with FastAPI, containerized releases, trace IDs, retries and instrumented operational dashboards.
- Ran incident reviews for incorrect model claims; added structured output validation and cross-document authorization tests.
- Designed source-aware prompt templates and evaluation cases that covered missing documents, low-quality retrieval and contradictory passages.
- Added a model-provider fallback path with strict output schemas and safe error-to-review behavior for malformed replies.
- Analyzed retrieval ranking and annotation disagreements with domain reviewers, then prepared traceable fixes across releases.
- Created API integration tests that blocked answers citing documents outside the current customer account.

NLP / ML Engineer | Invenio Systems (fictional) | **Oct 2022 - Sep 2024**

- Trained PyTorch transformer classifiers for automated document tagging using annotated multilingual examples and stratified test data.
- Created NER and information-extraction workflows supporting service operations, with explicit error analysis by document type.
- Built batch demand-anomaly screening using Python, seasonal features, out-of-time validation and human feedback loops.
- Integrated internal search APIs and experiment tracking into web applications used by account teams.

ENGINEERING DEPTH AND IMPACT

METHODS & PRODUCTION WORKFLOWS

- Designed retrieval pipelines, indexing and rank evaluation using Haystack and direct Embedding APIs, with source tracing maintained in Python.
- Orchestrated LLM calls and tools using conditional Python workflows; separated functional steps to preserve reproducibility and error recovery.
- Worked with SME reviewers to define supported-answer criteria, abstention conditions and acceptable ambiguity ranges.
- Documented latency budgets, cache invalidation, failure retries and source-ownership checks for protected document collections.

SELECTED EVALUATION DETAILS

- Audited 450 manually judged policy answers with source IDs and reason codes; assessed retrieval miss rates independently of generation quality.
- Benchmarked PyTorch NER and classifier models using precision, recall, macro F1 and grouped error reports for unseen document layouts.
- Ran multi-prompt LLM experiments while preserving a hold-out set of evaluation cases and showing changes in hallucination frequency.
- Demonstrated functional transferability to chain/agent-style systems through custom state routing and Haystack pipeline composition.

TEAMWORK & PROFESSIONAL PRACTICE

- Built searchable incident reports for model failures and helped support teams separate missing documents from unsupported model claims.
- Produced API schemas, test fixtures and reviewer-facing examples for engineering hand-offs across retrieval and NLP services.
- Presented methodology to product owners and documented that keyword proximity is not proof of functional competence.
- Kept the distinction between research prototypes, production deployments and unverified tool familiarity in project documentation.

SELECTED PROJECT OUTCOMES

Knowledge-grounded response platform: Haystack + hosted LLM API; same retrieval, tool routing and evidence assembly functions often achieved with LangChain.

Conditional workflow engine: Python state graph with branching, tool retries and reviewer approval; transferable to graph-based orchestration concepts.

Multilingual invoice labeling: PyTorch transformer with traceable annotation runs and 88% macro F1 on independent evaluation split.

EDUCATION, METHODS AND ADDITIONAL BACKGROUND

- BS Computer Science | Harbor Technical College (fictional), 2022
- Has not used LangChain or LangGraph professionally; has built comparable retrieval and workflow functionality using other tools
- Quality responsibilities: adjudication rubrics, small-dataset error review, response abstention, observability
- Stakeholder communication: acceptance criteria, evidence reviews, documentation for support and product teams